

AJAY SUSANTH

SOFTWARE ENGINEER

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PROFESSIONAL SUMMARY

Final-year B.Tech student with hands-on experience building and operating production-style systems end to end — from cloud infrastructure to running services. Designed and provisioned Azure infrastructure with Terraform, built distributed backend systems with FastAPI and Express.js, and re-architected a monolithic deployment into a scalable, observable Kubernetes platform with CI/CD and GitOps. Adept at owning a project from infrastructure to production, not just writing application code.

SKILLS

Technical Skills: Python, MERN, PostgreSQL, Docker, Kubernetes, GitHub Actions, Azure, Terraform, Postman, n8n automation.

Soft Skills: Problem solving, Leadership, Teamwork, Communication, Adaptability, Attention to detail, Time management.

PROJECTS

SCALABLE N8N DEPLOYMENT ARCHITECTURE | AKS, Kubernetes, Terraform, ArgoCD, Prometheus

[GitHub](#)

- Redesigned n8n's monolithic deployment into independently scalable UI, webhook, and worker workloads on AKS, provisioning infrastructure through reusable Terraform modules.
- Built a GitOps deployment workflow using GitHub Actions, Helm, and ArgoCD, enabling automated Kubernetes releases with immutable 1.0.0-<SHA> versioning.
- Implemented KEDA event-driven autoscaling, scaling worker pods from 1–10 replicas based on Redis queue depth, with Prometheus and Grafana dashboards for operational visibility.
- Strengthened cluster security using 3 OPA Gatekeeper policies, Azure Key Vault CSI integration, and Calico Network Policies enforcing default-deny traffic controls.

TALENTFORGE: SMART TALENT SELECTION ENGINE | FastAPI, React, PostgreSQL, Gemini, Redis

[GitHub](#)

- Built an async resume ingestion pipeline using FastAPI, Celery, and Redis with MIME validation, 10 MB upload limits, and idempotent task orchestration for reliable background processing.
- Developed a multi-stage LLM parsing pipeline integrating Azure Document Intelligence, HuggingFace NER, and Gemini/Groq fallback models to generate structured candidate profiles and normalized skills data.
- Architected a hybrid ranking engine using 1536-dim OpenAI embeddings and configurable rule-based scoring (40% semantic / 60% rules), backed by pgvector with an HNSW index for cosine similarity search at scale.
- Designed an LLM explainability layer that asynchronously generates ranked-candidate rationales (Gemini primary, Groq fallback), cached in Redis with a 7-day TTL to eliminate redundant inference calls.

CLASSSYNC: CLASSWORK AUTOMATION PLATFORM | Express.js, n8n, Terraform, Azure, PostgreSQL

[GitHub](#)

- Architected a multi-tenant REST API using Express.js with classroom-scoped authorization for assignment and grading workflows
- Integrated n8n-based workflow automation with webhook-driven event processing and authenticated internal APIs for notification delivery and reminder scheduling.
- Provisioned Azure infrastructure (Container Apps, ACR, PostgreSQL Flexible Server, Log Analytics, Key Vault) via Terraform, enabling repeatable IaC deployments with centralized secret management.
- Containerized with multi-stage Docker builds behind a NGINX reverse proxy; designed a Telegram-first account-linking flow using short-lived JWT tokens and chat-ID binding.

EDUCATION

APJ Abdul Kalam Technological University

2023-27

Mar Athanasius College of Engineering

Kothamangalam, Kerala, India

Bachelor of Technology in Artificial Intelligence and Machine Learning

8.78 CGPA

CERTIFICATIONS

- Oracle Cloud Infrastructure Certified Foundations Associate, 2025
- Deploy Kubernetes Applications on Google Cloud (Skill Badge), 2025